

Part Three

FACTORS THAT INFLUENCE TRAINING

ENDURANCE, CARDIO, CROSS TRAINING, HYBRID TEMPLATES, AND ROUTINES

Not all athletes will be solely focused on bodyweight training as their primary form of strength work. Here are several factors that will influence your training, no matter what your goals.

ENDURANCE AND CARDIO

Endurance is much easier to understand than strength training. It is a catch-all term to describe high repetitions of any particular activity. While in fact it is distinct from metabolic conditioning and circuit training, we often use the term *endurance* to encompass all of these concepts.

Cycling, long distance running, long distance swimming, and other 10-20+ minute sustained workouts are typically called *cardiovascular endurance activities*. Metabolic conditioning is generally used in the context of training the various metabolism systems that produce energy in your body. Typically, this type of training involves different types of activities performed in a sustained manner over time. The training usually takes place in single-digit minutes, but can often last longer. Circuit training typically involves moving from one weight lifting exercise to another, and although it is possible to involve disparate activities, usually its practitioners exclusively focus on weights.

It is important to know about energy pathways. Phosphocreatine or creatine phosphate provides most of the energy for activities with very short durations (0-10s), the glycolytic pathway provides most of the energy for activities with moderate durations (10-75s), and the oxidative phosphorylation pathway provides most of the energy for activities with longer durations (75s and longer). Phosphocreatine and glycolytic systems are termed *anaerobic* and the oxidative system is termed *aerobic*. Here is summary of Table II from Gastin's review on energy system interaction and relative contribution during maximal exercise:

Duration of max exercise	% Anaerobic	% Aerobic
0-10 seconds	94	6
0-15 seconds	88	12
0-20 seconds	82	18
0-30 seconds	73	27
0-45 seconds	63	37
0-60 seconds	55	45
0-75 seconds	49	51
0-90 seconds	44	56
0-120 seconds	37	63
0-180 seconds	27	73
0-240 seconds	21	79

Studies on elite track and field athletes tell a similar tale. For 400m athletes, where the men's world record is around 43 seconds, the contribution is 60/40 anaerobic to aerobic. For 800m athletes, where the men's world record is around 100 seconds, the contribution rises to 40/60 anaerobic to aerobic. This agrees with the 1600m world record of 223 seconds, which is approximately 20/80 anaerobic to aerobic.

This indicates that if you want to train cardiovascular endurance capacity, your training needs to be significantly longer than four minutes in order to work the capacity of these systems. Studies and training programs show that to work these systems well, sub-anaerobic threshold running is recommended for periods of twenty minutes to an hour or more.

In terms of any type of "full energy system training" or metabolic conditioning, there are programs like *CrossFit*. In most of the short-duration CrossFit workouts (which run two to five minutes) energy is primarily provided by your aerobic system. To truly work the anaerobic system to maximal capacity, you should undertake 30-75 seconds to minimize both phosphocreatine and aerobic contribution. Anecdotally, this is the time when metabolic acidosis ("the burn") is primarily felt—which coincides with scientific research. If you analyze CrossFit workouts, you will find that there is a lot of overlap and need to be efficient in all of the pathways.

Both endurance and metabolic conditioning work similarly on a biological level. High repetitions for endurance require a high level of neural adaptation in the specific technique as well as a high level of energy pathway efficiency in order to continually produce energy. Metabolic conditioning requires the exact same adaptations, with a slight lessening of CNS adaptation, leaning more toward very high energy pathway efficiency since there are more exercises to perform. (Hence, metabolic conditioning is somewhat lighter on the CNS or central nervous system.) One of the most important attributes of these workouts—aside from reaching your goals—is they both work toward increasing your overall work capacity. This means that if you suddenly decided to begin strength training—even if you are combining strength training with endurance or metabolic conditioning—your muscles and CNS should be able to handle more work while you are performing a routine. This allows you to stress them more for increased hypertrophy without overtraining.

Adding endurance into a routine follows the same guidelines as strength training without the exceptions. It is acceptable and even encouraged to go to failure most or all of the time. This is especially true with metabolic conditioning. You can go to failure all of the time here because the many different exercises performed

have a high energy requirement, which puts stress on your energy systems and relatively less on the nervous system. Therefore, it will not burn out your nervous system and will still stimulate your muscles enough that they adapt and produce energy more efficiently. This in turn increases your work capacity next time you exercise. Pushing “through the burn” here results in raising your lactic acid threshold (glycolytic pathway). On the other hand, not going to failure can also be utilized well with endurance and metabolic conditioning. In metabolic conditioning, if you stop short of failure, your work capacity will increase the next time you exercise within the same session. This means you will be able to perform more work in a shorter amount of time, and your total power output will rise. This is akin to a higher intensity exercise, because although it is different from a 1 RM exercise, the total intensity is distributed throughout your body. In this one instance, metabolic conditioning is more similar to strength than endurance.

Now that you have a foundation in energy systems, we can discuss some examples of how this knowledge might be used in your workouts.

To-Failure Training

Obviously, one of the ways to perform endurance exercises is to push it to the max. Failure is optional, but not encouraged on the first set because it decreases your maxes for subsequent sets. The total number of repetitions for the whole workout will usually be higher if failure is not hit on the first few sets. For example, if you can perform a maximum of twenty dips and want to increase to fifty dips, then one endurance workout could be 3x18. If you only performed fifteen, then 5x15 would be a possibility. Keep in mind that the higher number you pick, the fewer sets you will be able to perform. Generally, the best workout is one that maximizes the number of repetitions per workout, so 5x15 would be better than 3x18 since it is seventy-five repetitions vs. fifty-four. Some people are naturally inclined to endurance exercises, having greater proportions of slow-twitch muscle fibers. They might be able to perform much closer to their endurance RM for more sets than other athletes. This is true of women compared to men. Find out what your body can do and go with that.

Grease the Groove

Grease the groove can be used for strength or modified for endurance. Here is the way it was performed for strength training: Six to ten submaximal sets or more, interspersed throughout the day. Typically, you will perform about 60-80% of your maximum number of repetitions. We will use dips as an example. If your maximum is four dips, you will want to perform about two or three dips for one set each. You will then perform dips six to ten times every hour or two throughout the day. This leads to performing up to thirty dips during the day, whereas if you were to perform dips only during your workout you would probably be able to do only four or five sets of three before you become fatigued and unable to do more. If we continue with the four dips example, it may only take a few weeks for you to develop the ability to perform ten or more. Staying submaximal with your sets will help you avoid overworking your body, but gives you enough practice to make you quite good at the movement in a short period of time.

To modify for endurance, simply use an exercise that you can perform many repetitions of and modify it accordingly. Thus, if you can perform a max of twenty pushups in a set, you will perform 60-80% of that number (six to ten sets of twelve to sixteen pushups) throughout the course of a day. Since endurance work is less intense than strength work, you can likely move up to 75-90% to work sets of fifteen to eighteen repetitions. Going to-failure or near-failure is needed in order to maximize endurance gains.

Ladders and Pyramids

Ladders are a fairly simple concept. You start a timer and do one repetition of an exercise and then rest until the minute is up. Next, you do two repetitions of the exercise and rest until the minute is up. Continue this progression until you cannot complete the ladder any longer. A minute is an arbitrary amount of time; you can use any amount of time you desire. You also do not need to go up the ladder in a “one, two, three” manner. Instead, you could begin at five and go “five, ten, fifteen” or any variation thereof.

Pyramids are similar to ladders except after you “climb the ladder to failure” you also climb back down. For example, if you completed seven repetitions of pull-ups and failed on the next set, you would “climb down the ladder” by performing repetitions of six, five, four, three, two, and one pull-up. As with ladders, you can go in increments of two’s, five’s, and so on.

Both ladders and pyramids are ways to drastically increase the volume of the number of repetitions of an exercise in a single session. If you take the example of the maximum of twenty dips in the to-failure training section, you might only work 5x15 or seventy-five total repetitions, with the last set to failure. If you contrast this to pyramid training, you are likely to work up to a maximal set of twelve repetitions ($1+2+3+4+...+10+11+12+11+10+...+3+2+1$). However, you will perform a total amount of 144 repetitions, nearly double the volume of 5x15. Much like hypertrophy, volume plays a big factor in obtaining endurance.

Metabolic Conditioning and circuits

Metabolic conditioning workouts are not difficult to construct, but it takes a good knowledge of your body to program them effectively. CrossFit is one of the most popular commercially-available types of “no rest” workouts, but you can choose two to five exercises that you normally perform more than ten repetitions of. Arrange them as you desire and set an arbitrary number of repetitions to perform for each exercise, usually somewhere between 25-75% of your maximum ability. Then, choose a number of rounds to perform the circuit, say three to five. Perform this workout as quickly as possible. Alternatively, there is the “AMRAP” method, which is “as many rounds as possible” in a particular time period.

Programming metabolic workouts is really an art, and you must have a specific goal going in to them. It is important to know whether you are attempting to train oxidative capacities or glycolytic capacities (aerobic or anaerobic energy pathways). Additionally, it is important to program a workout according to personal ability levels. Getting near the end of your workout and being so fatigued that you can only grind out single repetitions is not a particularly effective method for increasing capacity.

There is a thoughtless criticism known as “any asshole”: any asshole could throw together a bunch of exercises into a workout. Usually anyone who spouts the “any asshole” criticism also has the unconscious goal to feel miserable during their own workouts, rather than having a clear goal about how to develop their athletic capacity to perform better. While “any asshole” workouts can improve capacity to some extent, they will not improve it optimally. The practice leads to workouts that provide too much of a stimulus or devolve toward overtraining. More is not always better. Effective workouts increase work capacity both in potential metabolic conditioning and strength/endurance. A measurable amount of capacity should increase from workout to workout, week to week, or perhaps with repeat workouts.

High-Intensity Interval Training

High-intensity interval training (HIIT) and similar techniques, like fartlek, are good ways to work all of the energy systems in your body because it depletes them very fast. HIIT workouts are generally composed of all-out sprints for a short period of time (15-30 seconds), followed by jogging, walking, or another form of rest for 30-45 seconds. HIIT builds both the aerobic and anaerobic pathways in the body very quickly, which leads to good metabolic conditioning and cardiovascular health.

Training anaerobic vs. aerobic capacities is a matter of intensity and adjustment of volume. Anaerobic qualities, much like strength, are best trained by operating at maximal capacities with longer rest times. For example, you might sprint for ten seconds and rest for two to three minutes to work maximal anaerobic qualities. This ensures that the muscles are fully replenished with energy and that your next effort will be closer to maximal than if you had simply rested for only one minute.

Tabata protocol is a specific form of HIIT, named for the Japanese scientist who experimented with it by cycling for eight periods that consisted of twenty seconds of all-out activity followed by ten seconds of rest. This protocol is similar to metabolic conditioning and HIIT-type workouts, and it can be used to rapidly increase anaerobic capacity. Tabata protocol is very similar to HIIT except it is done with exercises other than running and jogging.

You can modify the Tabata protocol according to your needs. You could run on thirty or sixty-second intervals where you go all-out and then rest. For example, if you performed a thirty-second interval of bodyweight squats, you would do as many squats as you can for twenty seconds and rest for ten seconds. Repeat this process multiple times, usually five rounds or more. The sixty-second protocol works on the scale of forty-five seconds on and fifteen-seconds off. It is essentially a variation of a metabolic conditioning workout that works all of the body's energy systems very rapidly.

Cardio – Generalized Aerobic Capacity

Despite the fact that *low-intensity steady state* (LISS) endurance training is demonized by many, it remains the most effective way to increase aerobic capacity. There is a reason why every intermediate and long-distance runner uses LISS training. This type of endurance training is performed at or slightly below (5-10%) anaerobic threshold. Therefore the anaerobic pathways are not taxed, while maximal stress is put on the aerobic system in order to stress adaptations. To put this in a more colloquial way: stay slightly “below the burn” or at a low enough intensity that you can carry on a conversation while performing the aerobic exercise. Perform for thirty to sixty minutes or more, depending on your sport and/or the qualities you want to improve.

The main reason to place LISS training in a routine is it can improve recovery for strength workouts when it is performed at lower levels.

Should you or should you not perform extra cardio? What about other exercises or sports that are similar to cardio? (Pick-up basketball, football, and/or other sports.) To decide, you need to learn a few terms:

- **Passive Rest** – Doing absolutely nothing related to exercise or activity on a rest day. Passive rest days usually occur the day after a workout.

- **Active Rest** – Performing activities that get you moving without heavy exertion. In other words, staying below your anaerobic threshold while keeping active. Usually this occurs in the form of recreational sports.
- **Active Recovery** – Performing a mindful workout at a much lower intensity. This could be technique work like handstands or sport-specific technique work performed at low-intensity thresholds.

A common problem occurs in young and/or competitive athletes: both active rest and active recovery seem to always turn into a full-blown workout. If you are unable to remain low-intensity with your exercises, you should absolutely not perform any type of active rest or active recovery. Your body requires rest! Rest causes your body to get stronger and prepare for your next intense workout.

However, if you have the self-control to keep things low-intensity, you will usually find that some type of light work may help your training, especially as you progress in strength and work capacity. Indeed, it has been noted that many elite athletes that choose mostly anaerobic training yield benefits from low-intensity cardio work. The Chinese gymnastics team chooses this practice, as do some of the world's top powerlifters. This is why fighters and multi-round event athletes do significantly better with their “wind”—aerobic base—when they have LISS training in their program.

It goes without saying that low-level activity is generally good, as it increases parasympathetic nervous system stimulation which facilitates digestion and nutrient shuttling to repair muscles and connective tissues. It also increases blood flow to the brain, which improves mood and helps provide nutrients to recover neural stress. It can improve posture. It lowers resting heart rate. It increases work capacity during strength workouts through more efficient energy recovery.

Cardio work makes the most impact at intermediate and advanced ability levels. For beginners who are focused mainly on strength and hypertrophy, cardio does not need to be trained. You can improve without it. Rest days should be passive rest days unless you absolutely want to train cardio as it provides additional benefits. As you move into the intermediate and advanced range, some form of low-level activity (cardio, active rest, active recovery) may be useful, even on your rest days. This will promote an increase in overall work capacity and speed up overall recovery. As you become stronger, it will become more important for recovery.

Circus arts have active rest built in, with their extra technique training every day. They perform three to four hours or more of primarily non-strength technique training and get a lot of low-level activity to promote recovery. That is how these athletes are able to build up to working out five to seven days a week, often for six to eight hours at a time (with breaks). Even if you are not interested in honing your skills to acrobat levels, you can enjoy life while adding in active rest days using pickup sports or recreational swimming, cycling, jogging, hiking, or walking. All of these are great for recovery, so long as they do not become workouts themselves.

In studying how cardio affects elite-level athletes, we can draw the conclusion that activity is good for your body. As long as it is kept at lower intensities outside of workouts, it will help both your recovery and strength workouts.

There is often a concern that LISS cardio will make you slower, but that is simply not the case as long as you are training strength and power as primary attributes. Cardio simply becomes a small component of the

overall workout. We see the reverse of this in elite endurance athletes. They will typically train LISS anywhere from 60-80% of their workouts, but they will also aim at increasing strength, power, and anaerobic capacity, especially close to competitions. Strength, power, and anaerobic capacity can help endurance athletes win races in the final sprint; light activity can help strength and power athletes in recovery and work capacity.

All you need is a good balance. The *Pareto Principle* (or *80/20 Rule*) tends to be a good rule of thumb for maintaining a balance between power, strength, and endurance in your workouts. If you are a strength and power athlete, 80% of your training should be devoted toward that, with 20% of your training devoted to walking or LISS cardio to promote recovery—and vice versa for endurance athletes.

Whatever you do: *listen to your body*. This cannot be stressed enough. You may start out with the 80/20 Rule for strength and power vs. cardio, but if you are feeling particularly fatigued you may need to cut back on the strength and power work for a bit and increase recovery. If you sense that increasing cardio has negatively impacted your training, scale it back and focus more on strength and power. There is variance with any large population; adjust according to how your body responds.

It goes without saying that passive rest is not good for the body. There are studies that show those with desk jobs or who are inactive for long periods of time have a higher mortality rate, even if they exercise regularly. The body does not like inactivity, so if your passive rest is akin to sitting on the couch all day, that is not good. Going for a walk or light jog would be superior from a health standpoint.

CROSS TRAINING

You cannot combine a bodyweight program with another sport's strength and conditioning program and expect to have good results. Splitting time between activities will cause your gains to lag in each if you are not a beginner. You also have a greater chance of overtraining. Therefore, if you have significant aspirations in other sports, it may be a good idea to take a long look at bodyweight training with your coach to determine if it may negatively impact your performance there. Sometimes it will; other times, it will not. Occasionally, with sports like wrestling or martial arts, bodyweight strength training may be the best way to improve strength and conditioning.

A lot depends on the coach. Some coaches are “my way or the highway.” If your sport is really important to you, do it their way. Other coaches are more flexible. They may still want you to do it their way, but if it's only strength and conditioning, and you have good information they may incorporate it. Some might not care at all about strength and conditioning, in which case the ball is in your court. Figure out what type of coach you have and act accordingly.

Ideally, your coach has your best interests at heart and you can trust them. Keep them in the loop if you intend to add or substitute bodyweight strength training for any other strength and conditioning work or for traditional weight training. You might even ask for a trial period where you try bodyweight strength training on an experimental basis. Be sure to stick with it if your results are good. Your coach can and should advise you. They have substantially more experience than you do and have good reasons behind the methods they use. If their reasoning seems outdated, bring them scientific research or other support from top-level coaches. The best compromise is to improve on your own and help your coach become better too.

Properly run strength and conditioning for other sports may start interfering with your bodyweight workouts. If you are serious about your sport, specific training for that sport should be above all else. If training a bodyweight movement such as handstands interferes with training for your sport, it may need to be eliminated. You should train for your sport when you are fresh, not when your energy level is depleted from training for less important things first.

Likewise, sport-specific strength and conditioning takes precedence for bodyweight training. Sometimes you can fit a bodyweight workout here and there; other times, you cannot.

If you are training or practicing for your sport five times or more per week, it would probably not be prudent to add any extensive bodyweight strength and conditioning workouts. However, skill-based training may still apply if you desire to work on bodyweight skills such as handstand work.

If your sport has no specific strength and conditioning work, it may be useful to add bodyweight training in its place. However, consult your coach first to see if there is any sport-specific strength and conditioning work that they recommend.

More is not always better! When in doubt, you should err on the side of eliminating extracurricular workouts. Integrate some bodyweight training into your workout regime only if you have the time and are not overtraining or burning out. Judge this by taking a close look at how you perform in your sport. In some cases (such as rock climbing or wrestling), bodyweight training can be extremely effective. It all depends on your sport and coach.

Be smart. You cannot do too many things at once. You have to prioritize what you want most in life; otherwise, you will not be good at anything. When in doubt, seek advice from those more experienced than you. Do not be afraid to be humble. Accept that you do not know everything and can always learn more.

Let's examine some specific integration with other sports and disciplines.

Gymnastics, parkour, and climbing each consist of three distinct components. There is a psychological component, consisting of overcoming fear and being able to perform under pressure. There is a skill-based component where the practitioner needs to learn all of the specific skills. Finally, there is a physical preparation component, which typically includes strength and conditioning based on increasing the athlete's abilities toward that discipline.

Bodyweight strength training distinctly falls under the umbrella of strength and conditioning, and is a safe and methodical way to increase physical abilities. This is consistent with every major level of organized athletics. Gymnasts do not just practice their skills—they also perform large amounts of strength and conditioning. Track and field athletes do not just run—they spend time in the weight room improving their strength or explosiveness. Swimmers do not just swim—they spend time in the weight room to improve their power.

Here is the fallacy of the recreational athlete: Because they do not believe in structured strength and conditioning, they typically practice only their favorite activity. Physical preparation via strength and conditioning should always be a separate but integrated part of training. Most sports or disciplines use the weight room; however, bodyweight strength training can be used effectively as well. To be great at your activity, you cannot solely practice it and expect to improve optimally.

There is another fallacy that often appears in recreational athletes. It is the notion that strength and conditioning should mimic sports activities. Studies have proven this to be false. Throwing heavier balls to get stronger or more powerful for baseball does not improve your ability to throw a ball faster. Swinging a heavier bat does not improve bat speed and power. Likewise, for sprinting, performing single-leg weighted half squats does not necessarily result in faster sprinting. This is important to know, as it allows us to broadly apply strength and conditioning principles toward developing specific attributes rather than to mimic sport-specific movements. It is more important to select exercises that optimally improve overall strength (or to work on weak links), rather than to attempt to mimic sport-specific movements. This overall strength will then be useful to the sport-specific movement you bring onto the field.

There is an interesting phenomenon across various disciplines with beginner and intermediate athletes. As mentioned, a general 80/20 Rule applies for elite athletes in sport-specific training as well as other types of complementary training. For beginner and intermediates, there is a different 80/20 rule: this one of prioritizing the time invested in sport-specific skills (80% of training) and strengthening and conditioning (20% of training). Thus, there should be an 80:20 or 4:1 ratio of sport-specific training to other types of training. 80% of your time should be spent practicing sport-specific skills and 20% of your time should be spent on physical preparation via strength and conditioning, mobility work, injury proofing, and the like. This is not just to improve your physical ability, but to help prevent catastrophic or overuse injuries. In context with the previous 80/20 Rule: 80% of your time should be spent on sports specific skills, 16% (80% of 20%) of your time on strength and conditioning primary attributes of your sport, and 4% (20% of 20%) of your time on cross training attributes.

Therefore, for an emerging discipline like parkour or climbing, the amount of time split across weekly activities should be approximately an 80/20 split for beginners and intermediates. Rafe Kelley (formerly of *Parkour Visions*), and Ryan Ford (of *Apex Movement*) both espouse a split of three to four days of parkour-specific practice to two days of strength and conditioning work. A sample weekly structure of a split between parkour-skill work and strength work may look like this:

Monday: Parkour Skill Work
 Tuesday: Strength and Conditioning
 Wednesday: Rest Day
 Thursday: Parkour Skill Work
 Friday: Strength and Conditioning
 Saturday: Parkour Skill Work
 Sunday: Rest Day

Given that the parkour sessions last for two to four hours, and your strength and conditioning work may last one to two hours, when you add up the hours on discipline practice and strength and conditioning you will see approximately an 80:20 ratio. For most beginner and intermediate athletes (of any sport), the preferred schedule is a full-body routine performed two to three times per week (depending on how much skill work must be accomplished in each session) performed in combination with two full rest days each week. A modification of the above schedule to place most of the parkour skill work on the weekends may look like this:

Monday: Full-Body Strength Work
 Tuesday: Rest Day

Wednesday: Parkour Skill Work
 Thursday: Full-Body Strength Work
 Friday: Rest Day
 Saturday: Parkour Skill Work
 Sunday: Parkour Skill Work

For other sports, such as gymnastics, the strength and conditioning work will usually be placed at the end of each practice. Practices can last four to five hours and may take place four to five times per week. Two to three hours may be spent on skill work, and the rest of the time is spent on flexibility and mobility training, adding strength and conditioning at the end. A similar schedule may be employed for climbing: hit the gym for three or four sessions per week and aim for an hour of strength and conditioning after each climbing session. Each of these routines tends to be around the 4:1 ratio of sport-specific work to physical preparation. In every case, it is assumed that your sport-specific skill work will last 2-4 hours each time you practice it.

Monday: Climbing + Strength and Conditioning
 Tuesday: Rest Day
 Wednesday: Climbing + Optional Strength and Conditioning
 Thursday: Rest Day
 Friday: Climbing + Strength and Conditioning
 Saturday: Flex Day (could be spent climbing, recovery, or resting)
 Sunday: Rest Day

Climbing is an interesting discipline to address because unlike most other athletics, which tend to do well with a upper/lower split (because you can bias the leg-intensive, sport-specific training on one day, and lower-body strength and conditioning on other days) the opposite is true for climbing. Here is an example of how this may play out (again, sport-specific training will last 2-4 hours)

Monday: Climbing + Lower-Body Strength and Conditioning
 Tuesday: Upper-Body Strength and Conditioning
 Wednesday: Rest Day
 Thursday: Climbing
 Friday: Upper-Body Strength and Conditioning
 Saturday: Climbing + Lower-Body Strength and Conditioning
 Sunday: Rest Day

There are obviously many things that go into strength and conditioning beyond just exercising to increase strength. Flexibility, mobility, prehabilitation, isolation work, and more can be trained on rest days, training days, in warm-ups, or after a workout. You can mix and match to fit your schedule. For example, many people climb three days a week and perform yoga as flexibility and mobility work after their workouts. Then, on their off days, they perform mobility and strength and conditioning work. Here is an example of what this might look like:

Monday: Climbing + Yoga
 Tuesday: Strength and Conditioning
 Wednesday: Climbing + Mobility Work
 Thursday: Rest Day
 Friday: Climbing + Yoga

Saturday: Strength and Conditioning

Sunday: Rest Day

You are not limited to a specific routine based on your sport. It is important to identify your weaknesses and modify your routine accordingly. If you have difficulty with certain movements (such as heel hooks in climbing), you may need to perform more flexibility work or yoga over the course of a cycle in order to improve your abilities. If you are having difficulty with climb-ups onto walls in parkour, you may want to spend extra time focusing on your skill work, as well as adding additional strength work specifically for pull-ups, dips, and muscle-ups. Let's say your strength and flexibility is good, but you needed to improve hand strength for climbing. This is easily done: add two or three sessions of hang board, campus board, or system board work to your routine. Always identify the specific areas that need to be improved if you want to progress.

Strength work is the foundation of athletic development. If you develop strength first, many other attributes and sport-specific skills will develop optimally, such as: cardiovascular endurance, stamina, flexibility, power, speed, agility, coordination, balance, and accuracy. Because everything builds on strength, it is the most important attribute for beginner and intermediate athletes to develop.

This is true even in endurance-related sports. Elite marathoners can run a mile in four and a half minutes, twenty-six times in a row. It is not possible to run this fast if you do not possess a large amount of strength endurance. Even the mile and 1500-meter world record holder, Hicham El Guerrouj, had strength and power work in his workout regimen at least three to four times per week, and he was running each of those stretches upwards of nine times, depending on the phase of his training.

You cannot ignore physical preparation if you want to improve in your sport. Weight training is a good way to gain physical preparation, however, bodyweight strength training is an excellent alternative to weight training, especially in certain sports and/or disciplines.

HYBRID TEMPLATES

Hybrid templates are for those athletes who are interested in using both weights and bodyweight for training. It is good to include all possible tools in your toolbox. There is a time and place for every type of exercise, aside from those that are unreasonably dangerous.

Some barbell-focused people snub their nose at bodyweight exercise, criticizing it for being too easy or too high-repetition without understanding the depth of what is possible. There are also some bodyweight fanatics who claim that weights are inferior to bodyweight training for various reasons. Both of these fanatical positions are short-sighted. Bodyweight and barbells complement each other extremely well in most circumstances, and many high-level athletes of various sports utilize both in their training.

There are many different methods to integrate barbell or dumbbell exercises into your routine. Here we address only the most common: *substitution*, *complementing*, and *supplementing*.

Substitution

For a full-body routine, here is the typical exercise prescription: two upper-body pushing exercise, two upper-body pulling exercises, and one to two leg exercises. This will keep your body balanced and allow good

strength and mass progression. Often the two upper-body exercises (for both pushing and pulling) are further differentiated by the selection of one exercise that focuses on horizontal movement and one exercise that focuses on vertical movement. This helps to maintain structural balance. If you are also using all compound exercises, a routine like this can hit every muscle in the body, awarding gains to strength and hypertrophy all around. A typical barbell-specific routine may look like this:

- Upper Push: Bench Press + Press or Dips
- Upper Pull: Pull-ups + Bent-Over Rows
- Legs: Squat + Deadlift

A matching bodyweight strength routine would seek to replace each of these weighted or barbell exercises with their bodyweight equivalents according to plane of movement:

- Upper Push: A Pushup or Planche Progression + Dips or Handstand Pushup Variation
- Upper Pull: Pull-ups + Inverted Rows or Front Lever Pull-up
- Legs: Squats or Pistols + Glute Ham Raise/Hamstring Curl

If your goal is to increase both strength and hypertrophy, these routines would be performed with difficult exercises, typically in the five to twelve repetition range. As you progress with barbells, add weight. As you progress with bodyweight, add repetitions until you can move to the next progression. It's that easy. Substitute a specific barbell or bodyweight exercise with its counterpart. If you want to add handstand pushups to your barbell-focused routine, substitute them for the other vertical pressing movement—in this case, press or dips. Want to work on the planche? Substitute it for the bench press.

Complementing

If your goals are more focused, you may want to *complement* your exercises. For example, many people want to work on both bench press and the planche at the same time. This is fine. They complement each other well. The bench press is great for adding brute pressing strength and hypertrophy, while the planche is great for increasing upper-body control and strength. People will say: “just train planche” or “just train bench”—the benefits to training one exercise exclusively are centered on your ultimate goal. Do you want to achieve a certain level of proficiency in just one? Or might you want to be good at both? There is nothing wrong with either choice.

In a classic beginner routine, you would eliminate the vertical pushing exercises. While being specific is good, you need to be wary of structural balance until you become more focused. This could become problematic if you are focusing all of your work toward vertical pulling. Horizontal pulling focuses your muscle strength and hypertrophy toward the scapular retractors and posterior shoulder much more than the vertical pulling exercises. By eliminating one of these types of exercises in your routine you risk creating an imbalance in your shoulder over time, as most vertical pulling exercises hit the chest and lats strongly, which are shoulder internal rotators. Most pushing exercises emphasize shoulder internal rotation as well. This encourages the body to be in posture where it is hunched forward, which is bad for structural integrity, especially if you have a desk job or are in school for many hours a day. This problem can typically be balanced with horizontal pulling. If you want to improve your horizontal pushing to the extent that you decide to forgo other types of pushing movements, try to balance them out with enough horizontal pulling to offset imbalances.

In conclusion, be aware of the risk-to-benefit ratios. If you notice that an imbalance is developing, back off and move toward a more structured, all-around routine. Continue until the imbalances are corrected; then, you can return to a complementing structure.

Supplementing

Many effective and proven routines, such as Jim Wendler's *5/3/1 Program*, use *supplemental* bodyweight or barbell exercises to help bring up weak links, provide conditioning, or develop other attributes. 5/3/1 works one primary exercise, such as squat, deadlift, bench press, or overhead press. You then add supplemental *assistance work* to these lifts. For example, with the deadlift, your supplemental work would be focused on your weak links. If you have weak glutes, work some hip/hinge exercises, such as good mornings. If you have weak hamstrings, work straight leg deadlifts, glute ham raises, or a similar exercise. If you have a weak back, work reverse hyperextensions. For bench press, choose some other form of pressing that focuses on weaknesses, or all-around exercises like dips.

Bodyweight strength responds in a similar manner. Simply use a bodyweight exercise as your main lift and supplement it with weights to focus on your weaknesses, hypertrophy, or whatever you wish to develop. As a more advanced athlete, depending on your goals and weaknesses, this can be an effective method to progress. In many cases, you can use weights to help supplement bodyweight exercises to work on both the neural adaptations and strength development needed for hypertrophy. Also, if you are dealing with injuries, you can substitute some weighted exercises for bodyweight exercises without aggravating your injuries so that they can fully heal.

Consider the planche. There are many ways to use weighted exercises to supplement its development. If you are working on brute pressing strength and potential hypertrophy, the smith machine is a useful tool. Using an undergrip / supinated grip, position yourself so your hips are underneath the bar instead of your chest. This puts the bar in a planche-oriented position. Now, load up the barbell with weights and do modified planche pushups.

It is a useful exercise for numerous reasons. Since your body is straight, along the bench, you do not have to worry about reinforcing the poor technique that commonly occurs when people arch their backs to compensate while performing planche pushups. You also do not have to worry about hitting the correct pushup angles with your shoulders, as the machine keeps the bar properly aligned. It also gives you a way to track your strength: with progressive loading. You can see how your strength is developing by tracking how much weight you can push. As an added bonus, after a certain point, weights can be a better stimulus for hypertrophy than bodyweight exercises, particularly in the shoulders and chest.

Dumbbells are also effective for many advanced bodyweight exercises. Going back to the planche example, you can work on dumbbell planche pushups like with a smith machine: Simply keep your hands above your hips while performing the pushup movement. Additionally, training more advanced exercises like maltese or inverted cross is easy with dumbbells. A favorite exercises to condition the elbow and build strength and hypertrophy involves using a pair of dumbbells:

- Begin in a planche position and hold for two seconds.
- Move to the maltese position and hold for two seconds.
- Move back to the planche position and repeat this sequence three to five times.

Of course, training on rings is preferable. Dumbbells are not a perfect substitute for rings, but not everyone has access to rings.

Another benefit of using weights is that you can interchange the weights to be just heavy enough to work on building up the connective tissue without causing injury. This is especially true for the straight-arm exercises that are very tough on the elbows (such as back lever, planche, maltese, and inverted cross).

A lack of gym availability is probably the most common issue I have seen in people who are training both weights and bodyweight. They have access to a *globo gym* (a disparaging term for a big-name, corporate gym) that does not allow them to perform bodyweight exercises and they have no equipment at home to work on bodyweight exercises, either. Others train consistently at home on the weekends and can barely make it to the gym on a weekday, leaving very little time to devote to training. If this is the case for you, adjust your routine to fit your schedule from the beginning. Depending on your goals, you may even want to have two separate routines that work around your schedule. A typical barbell routine is a routine you are usually able to perform at a *globo gym*.

- Upper Push: Bench Press + Press or Dips
- Upper Pull: Pull-ups + Bent-Over Rows
- Legs: Squats + Deadlift

An emulated bodyweight strength routine would seek to replace each of these weighted or barbell exercises with their bodyweight equivalents according to plane of movement:

- Upper Push: A Pushup or Planche Progression + Dips or Handstand Pushup Variation
- Upper Pull: Pull-ups + Inverted Rows or Front Lever Pull-up
- Legs: Squats or Pistols + A Glute Ham Raise/Hamstring Curl

Obviously, you want to modify these toward your goals. It is possible to train ground-based bodyweight exercises (such as the planche) at a *globo gym*, so you could do planche on both days instead of, say, bench press on your gym day. Alternatively, you can utilize the gym equipment to work on specific goals like smith machine planche pushups or dumbbell planche pushups.

How you structure your routines really is up to you and should be based on your goals. Upper-body bodyweight exercises and lower-body barbell exercises are highly recommended and proven to work. However, if you are interested in mixing and matching different bodyweight and barbell exercises for your upper-body and lower-body, go for it. Here is a non-exhaustive list of exercises that you can mix and match:

Barbell Pushing Exercises

- Upper: Bench Press, Dumbbell Presses, Decline/Incline Bench, Military Press, Behind-the-Neck Press, Jerks, Push Press
- Lower: Goblet Squats, Front Squats, High-Bar Back Squats, Low-Bar Back Squats, Overhead Squats, Hack Squats, Lunges

Note: Upright Rows are not on this list because of the shoulder impingement risk factor.

Barbell Pulling Exercises

- Upper: One-Arm DB Rows, Seated Rows, Bent-Over Rows, Pendlay Rows, Face Pulls
- Lower: Deadlifts, Snatch, Clean/Power Clean, any Olympic Lift Variation from ground to shoulder or hang to shoulder

Bodyweight Pushing Exercises

- Upper: Handstand Pushups, Handstand Presses, Planche, Pushups, Clapping Pushups, Dips, Maltese, Inverted Cross
- Lower: Bodyweight Squats, Pistols/Single Leg Squat, Shrimp Squats

Bodyweight Pulling Exercises

- Upper: Pull-ups (two arms, uneven, one-arm, etc.), Clapping Pull-ups, Inverted Rows, Inverted Pull-ups, Back Lever, Front Lever, Iron Cross
- Lower: Leg Curls, King Deadlifts, Glute Ham Raises

Miscellaneous

- Combined Upper: Muscle-ups, Inverted Muscle-ups, Any transitional movement involving a bar or rings (above-to-below or below-to-above)
- Combined Lower: Sprinting, Stairs, Vertical Leaping, Broad Jumping, and many more

Exrx.net has these and many more exercises with animated gifs to show you how they should look. In general, the classification system for the legs gets a bit fuzzy, as most leg exercises work every leg muscle. The main distinction generally made is that of push vs. pull, but the another distinction is that of knee dominant vs. hip dominant or hip hinge. Make sure you have balance in your workouts and work on any weak points you may have.

ROUTINES

Routines, sequences, and combinations are highly underrated and are actually an interesting way to work the muscles. One of their primary benefits is they allow you to transition in and out of many different types of movements consecutively using your muscles in ways you would not with typical exercises. Transitions from exercise to exercise can be very difficult and thus beneficial for gaining great concentric range of motion strength.

Routines, sequences, and combinations are best suited to advanced level strengths. Strength is mostly neurological adaptations, so it is best to first train movement patterns in typical repetition exercises in order to build the requisite strength. Training multiple exercises and moving through different transitions is great, but it does not allow you to stay as focused on the movement patterns themselves. You can liken this to a barbell complex or quick circuit of different strength exercises—it works better once you are already fairly strong.

Routines or combinations can consist of any type of exercises strung together in a series. They are usually performed on rings or parallettes. There are literally thousands of different variations. Some of the most common combinations are listed in the progression charts from the FIG COP along with a few variations of other movement combinations like *handstand* → *elbow lever* → *handstand*. Sample programming can be found in Part 4. These combinations are a very good way to get in some quick work if you do not have time for regular exercises. You can throw together a sequence of five to six skills and perform it five times in a row for a decent training session that makes a nice switch from “the grind” (doing the same or similar routines repeatedly).

Routines are an excellent way to incorporate skill and strength moves that you have already mastered but have not used in a while. Combining mastered movements with movements you are currently working on can create some interesting routines/sequences/combinations that will work your muscles differently than your usual methods. This is one of the major ways advanced-level and especially elite-level strength is developed. At these levels, your focus starts to shift from obtaining higher-level isometric holds to moving between these skills. Transitions between isometric holds (such as from the front lever to iron cross) may actually be harder than the holds themselves. Front lever is an A-rated skill, iron cross is a B-rated skill, and front lever to iron cross transition is called a *Pineda*, which is a D-level skill. Indeed, many high-level gymnasts cease doing specific strength work and focus solely on the strength skills in their routines to provide enough stimulus for adaptation.

Though routines and sequences can be used successfully to work on all-around movement strength, it is important not to remove all of your dedicated strength work.

Routines, sequences, and combinations can be fun to perform (and film!) as well. Play around with these types of movements once you can perform many of the high-level isometric holds, especially on rings. You may discover that you enjoy performing these types of routines more than the strict training of particular movements!

Remember, there is some trade off between working specific movements in a dedicated fashion to gain strength vs. working routines, much like there is some trade off between doing full range of motion concentrics vs. isometrics. If it is a trade off you are willing to make for the sake of variety or for your goals, go for it.

Here is the “beginner routine” used as a standard by Gymkana, an exhibitional gymnastics group, for decades:

- Begin from Hang
- Muscle-up to Support
- L-Sit
- Shoulder Stand
- Come down to Support
- Immediately roll back into Inverted Hang
- Back Lever
- German Hang Pullout
- Swing to Flyaway Dismount (or move back to hang and drop off)

This is one example where many basic skills are put into a sequence. These can be fun, especially when training with other people.

CHAPTER 13 SUMMARY

ENDURANCE, CARDIO, CROSS TRAINING, HYBRID TEMPLATES, AND ROUTINES

In this chapter, we looked at how endurance, cardio, cross training, hybrid templates, and routines apply to bodyweight strength training workouts.

Endurance trains both aerobic endurance and local muscular endurance. There are multiple techniques that you can use to construct your workouts.

Cross training is also highly variable, and your ideal routine will depend on the sport, coach, and level of strength and conditioning program implemented. If you are interested in learning bodyweight strength progressions and skills, have a conversation with your coach first and err on the side of less over more.

Strength and conditioning are a fundamental aspect of various sports. Specifically, strength and conditioning are important for improving effectively and to prevent injury.

Hybrid templates can be used to mix and match exercises from barbell and bodyweight exercises into a single routine.

Routines have their place as an alternate training method, and can be useful for increasing performance, working movements in succession, or simply having fun.